Canonical Camera Space Transform — Image Space

CSTM$_{\textbf{image}}$ resizes the input image instead of scaling labels:

$$\boxed{I_c = \mathcal{T}(I,\ \omega_r), \qquad \omega_r = \frac{f^c}{f}}$$

Why resize? After resizing, objects in I_c appear at the angular size they would subtend through the canonical camera. The encoder sees a consistent "canonical viewpoint."

Both paths are equivalent:

- CSTM$_{\text{label}}$: scale depth labels, keep images
- CSTM$_{\text{image}}$: resize images, keep depth labels

In practice, the paper uses **CSTM**